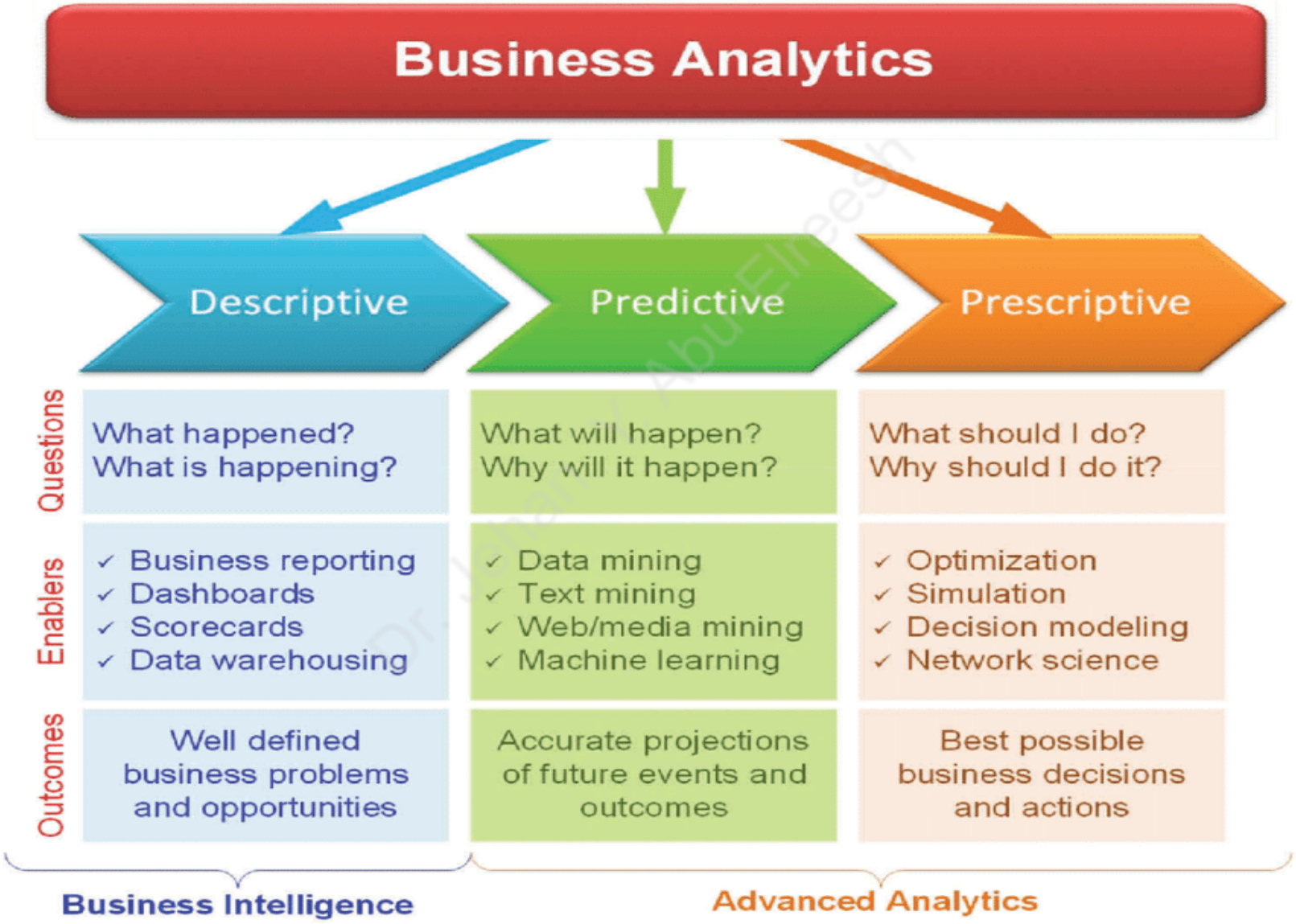
Data Warehousing and Modeling

Lecture 1
Introduction to Course

Dr. Jehan Y. Abu-Elreesh

2025 - 2026

Relationship between BA, BI and DW



Relationship between BA, BI and DW

Business Intelligence (BI)

Business Intelligence (BI) uses **Data Warehouses (DW)** as their foundation, with the **DW** acting as a central, organized data repository (the "what happened") that fuels BI's analytical tools (the "why it happened") for reporting, dashboards, and insights to drive better business decisions, while **Business Analysis (BA)** focuses on defining needs and bridging tech/business to ensure these systems solve real problems, often using data from the **DW**. In short: **DW** stores data, BI analyzes it for insights, and BA ensures the whole process meets business goals.

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Relationship between BA, BI and DW

Data Warehouse (DW)

Role: A central repository for storing large volumes of integrated data from various sources, structured for analysis.

Focus: Data storage, organization, and providing a single source of truth.

Analogy: The foundation or the "gold mine" of data.

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Relationship between BA, BI and DW

Business Intelligence (BI)

Role: The processes, tools, and technologies (like dashboards, reporting) used to analyze data from the DW.

Focus: Generating insights, trends, and performance metrics (e.g., sales, forecasts) to understand past and current performance.

Analogy: The "alchemy" that turns raw data into valuable "gold".

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Relationship between BA, BI and DW

Business Analyst (BA)

Role: Bridges the gap between business needs and technical solutions.

Focus: Gathering requirements, defining problems, ensuring the BI/DW solutions deliver real business value, and interpreting results.

Analogy: The architect or strategist ensuring the house (DW/BI) is built for its inhabitants (the business).

The Interconnected Relationship

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Relationship between BA, BI and DW

Data Collection: Raw data from operations feeds into the **DW**.

Centralization: The **DW** consolidates and cleans this data.

Analysis: **BI** tools query the **DW** to create reports, dashboards, and analyses.

Action: The **BA** helps interpret these **BI** outputs, ensuring they answer key business questions and drive strategy.

Essentially, **DW** is the *infrastructure*, **BI** is the *analysis*, and **BA** is the *strategic link* ensuring it all works to solve business problems.

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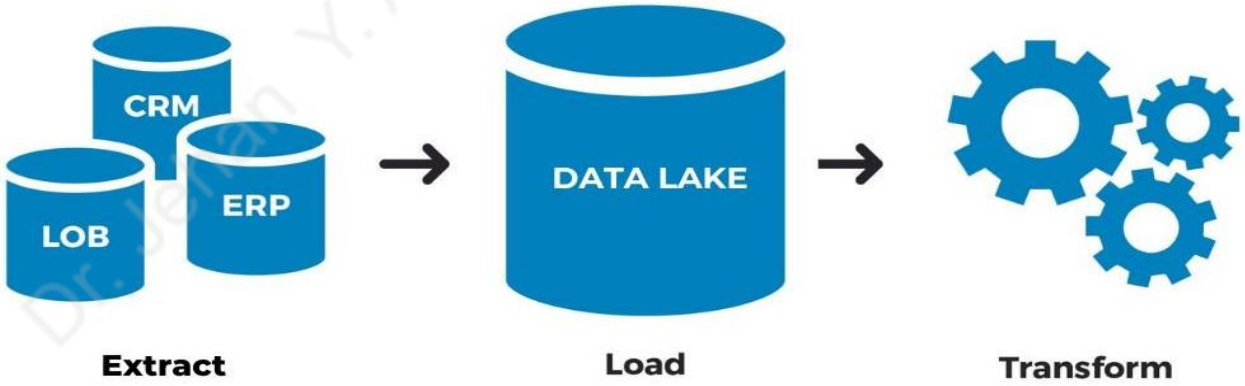
ETL vs. ELT



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ETL and ELT

ELT (Extract, Load and Transform):

Extract: Similar to ETL, data is initially extracted from source systems.

Load: In ELT, the data is first loaded into the target storage or data warehouse without significant transformation. It's stored in its raw or semi-structured form.

Transform: Transformation occurs after the data is loaded into the target system. Users can then perform transformations, data cleaning, and modeling using the processing power and tools available within the target system, often a data warehouse like Snowflake or Google BigQuery. ELT is more suitable for handling unstructured or semi-structured data and is commonly used in modern data warehousing architectures.

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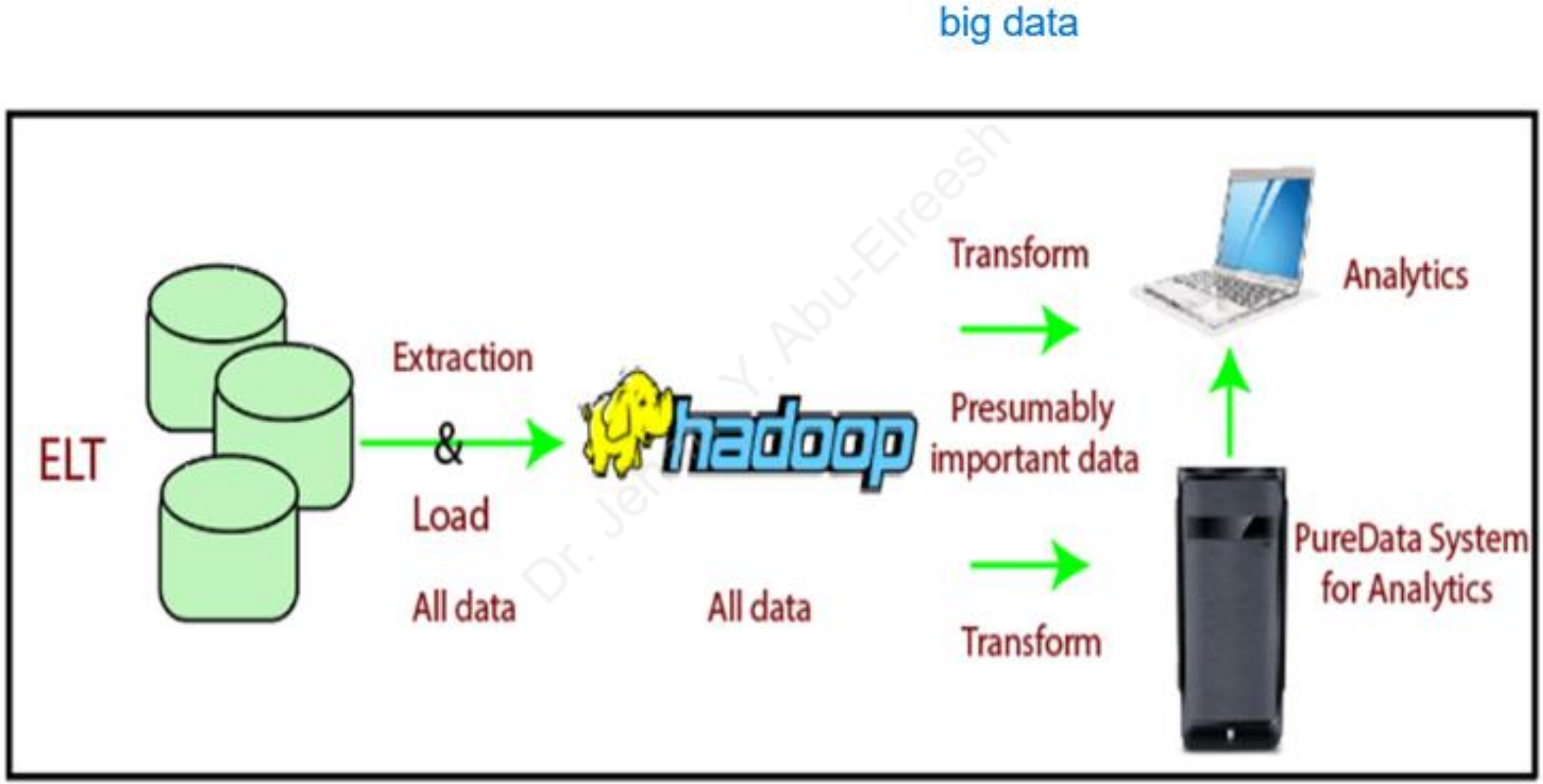
ELT (Extract, Load and Transform)



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ETL and ELT

ETL (Extract, Transform and Load):

Extract: In the ETL process, data is first extracted from various source systems such as databases, files, or APIs.

Transform: After extraction, the data undergoes a transformation phase where it is cleaned, filtered, and manipulated to meet the specific requirements of the target data warehouse or analytics system. This may involve data cleansing, aggregation, data type conversion, and other transformations.

Load: Finally, the transformed data is loaded into the target data warehouse or repository, where it can be accessed and analyzed by business intelligence tools, reporting systems, or data analysts.

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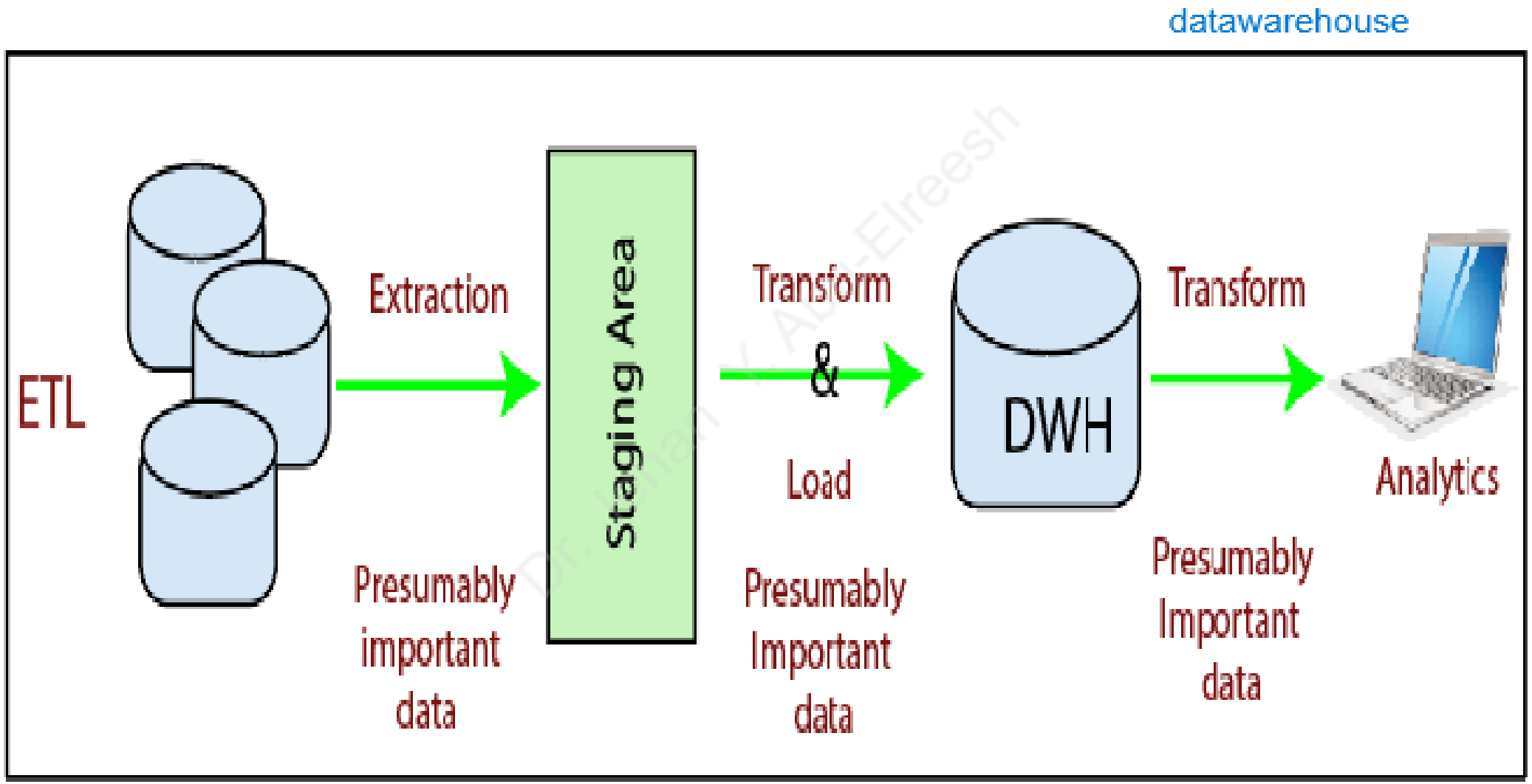
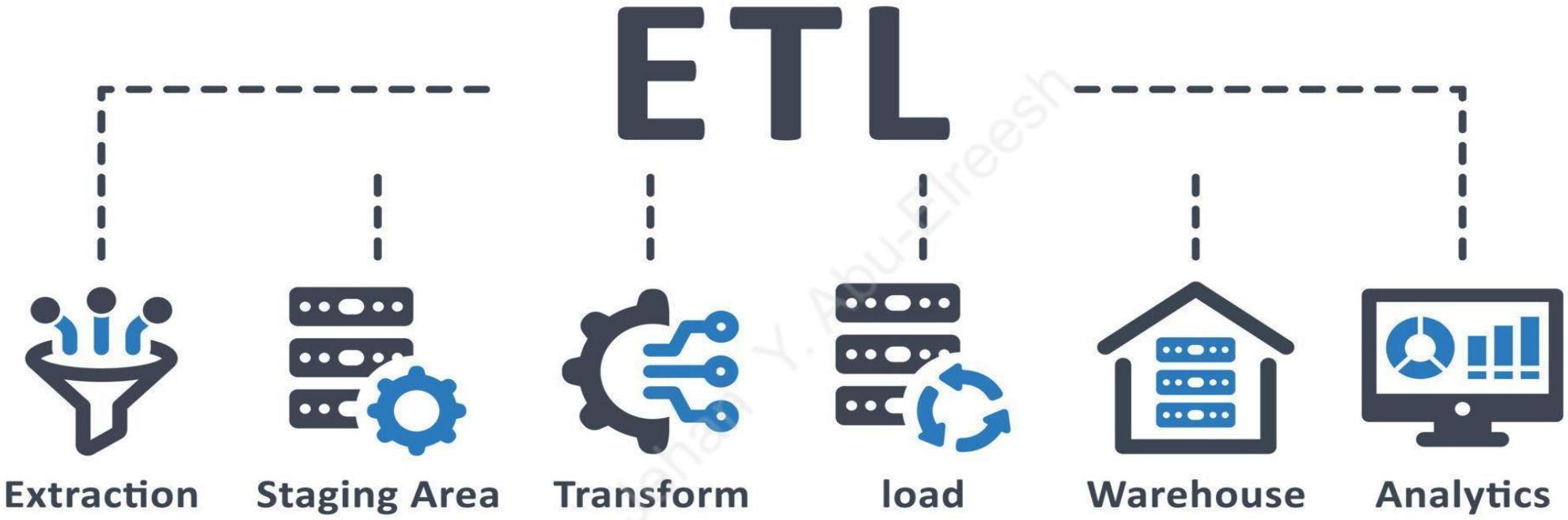
ETL (Extract, Transform and Load)



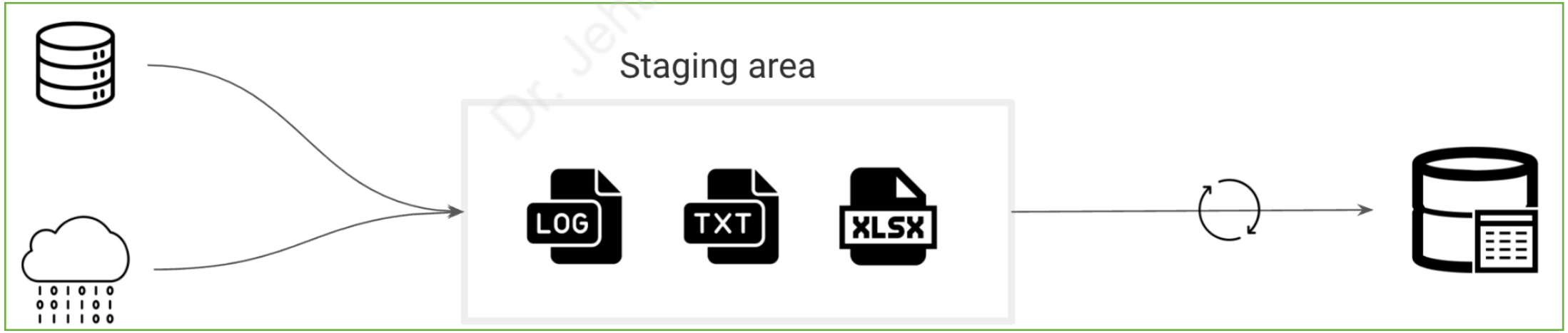


ETL is often batch-oriented and tends to be used for structured data.

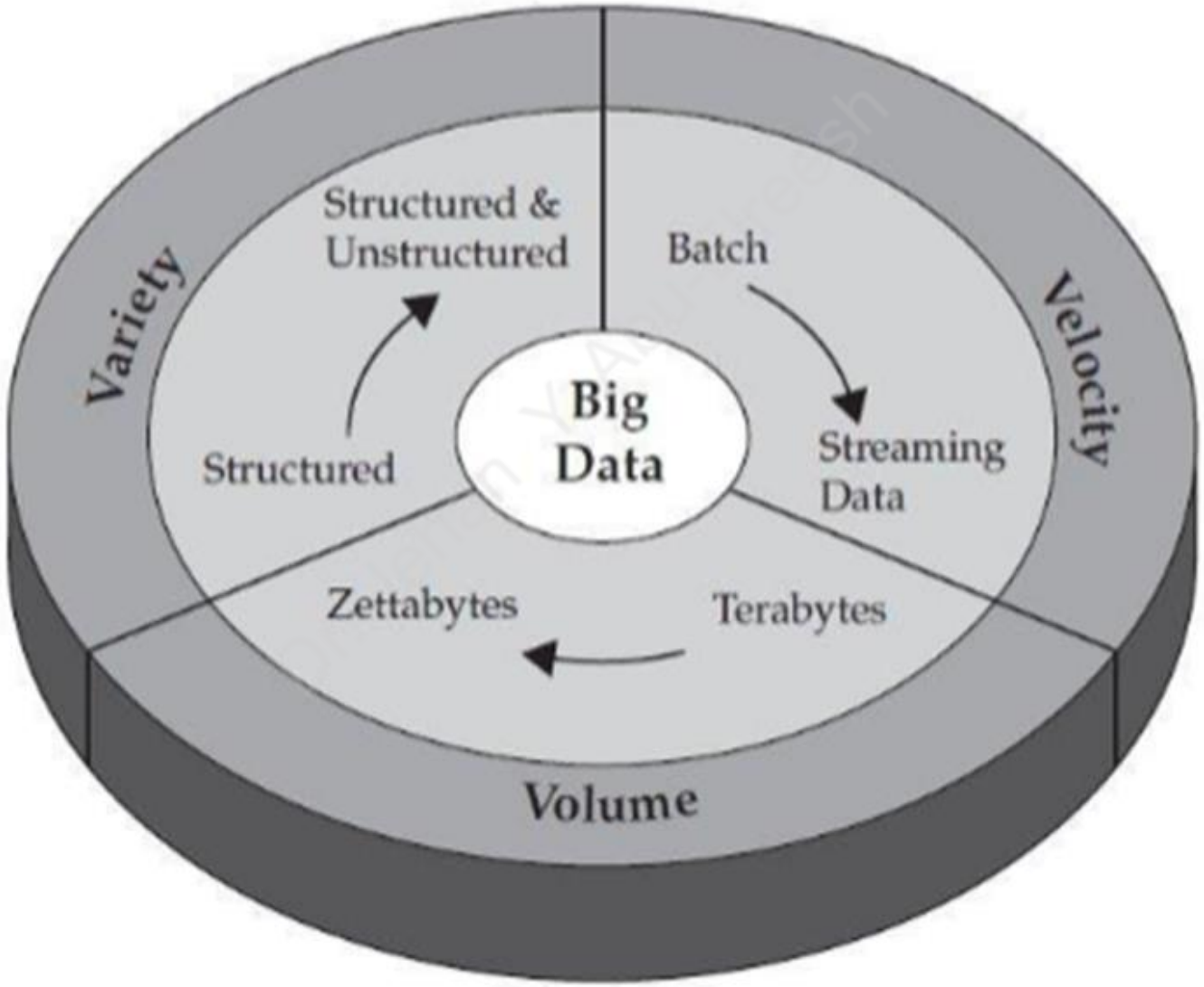
Staging Area

A staging area in data engineering:
is a temporary storage area where raw or unprocessed data from various sources is initially loaded and stored before it undergoes further processing, cleansing, and transformation.

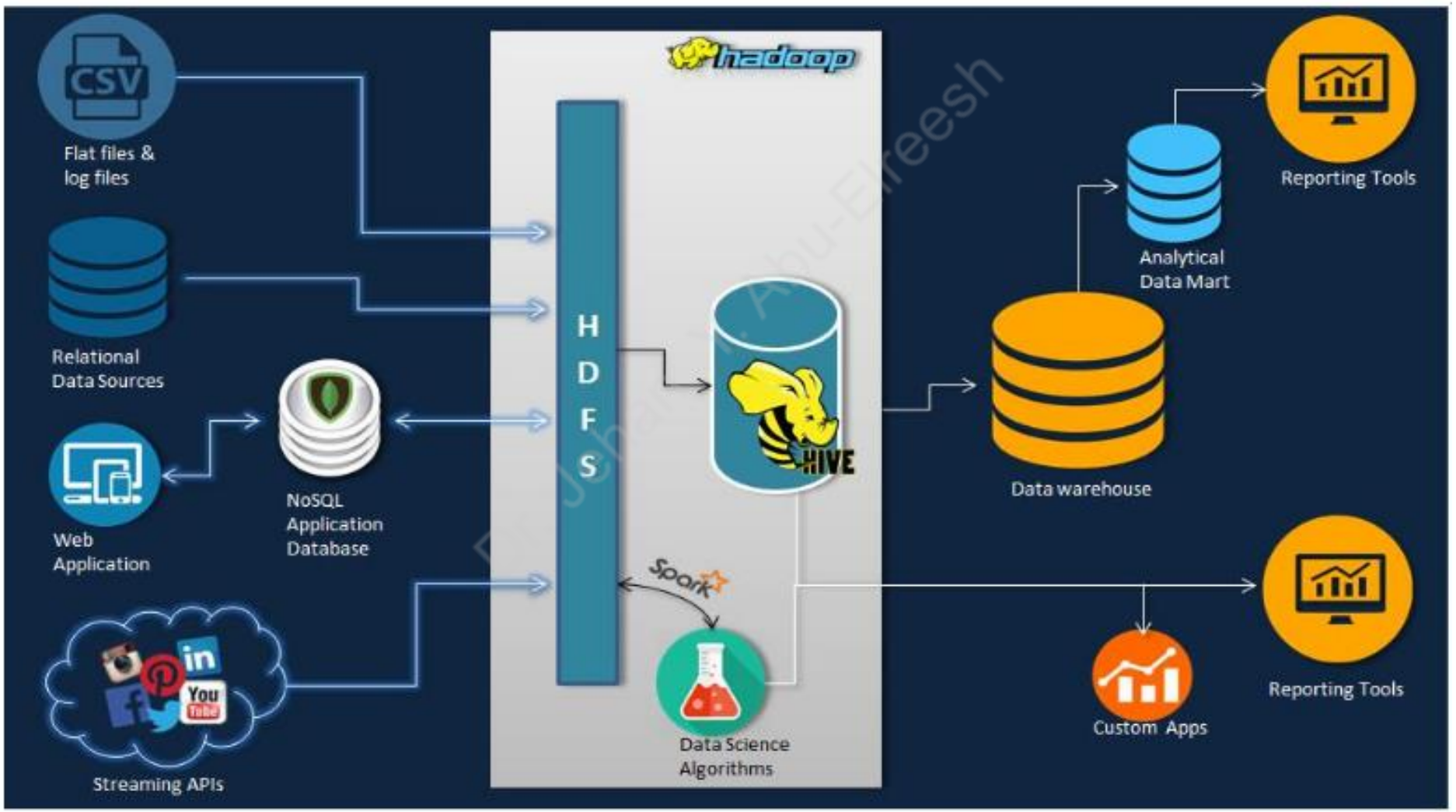
- ✓ It serves as a buffer between source systems and the data warehouse or analytics platform, allowing data engineers to validate, clean, and prepare the data for analysis without affecting the performance or integrity of the target system.
- ✓ Staging areas are crucial for data quality control and ensuring that only reliable and well-structured data enters the final storage or analytical environment.

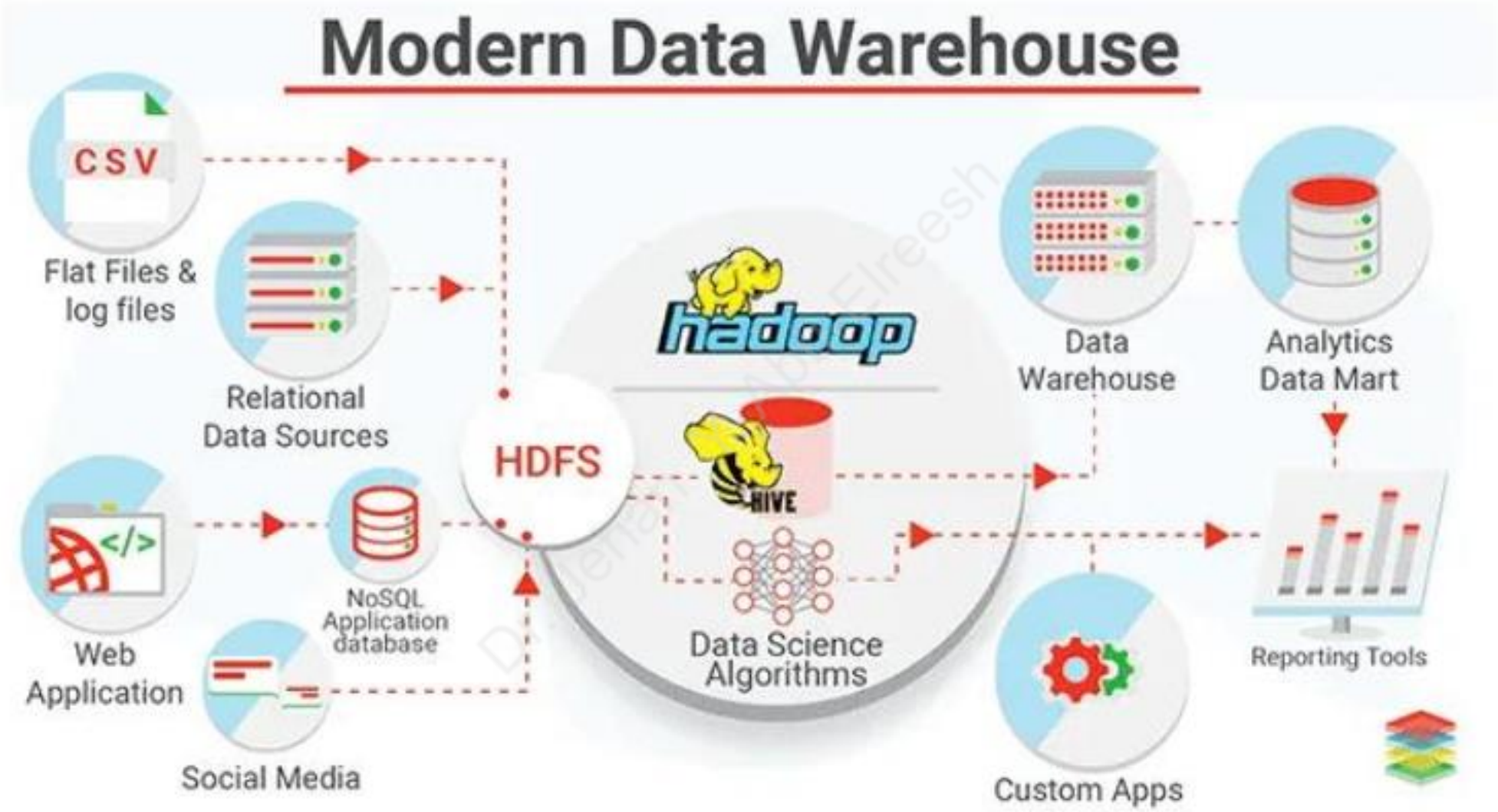


Bigdata


Variety
Structured &
Unstructured
Batch
Velocity
Big
Data
Structured
Streaming
Data
Zettabytes
Terabytes
Volume

Data Warehouse and Big Data




Modern Data Warehouse
CSV
Flat Files & log files
Relational Data Sources
Web Application
NoSQL Application database
Social Media
HDFS
hadoop
HIVE
Data Science Algorithms
Data Warehouse
Analytics Data Mart
Reporting Tools
Custom Apps

Data lake vs Data Warehouse

Data Lake:

Suitable for storing raw, diverse data types.
Schema-on-read for flexibility.
Supports batch and real-time processing.
Ideal for big data and exploratory analysis.
Cost-effective for large unprocessed data.

Data Warehouse:

Designed for structured data with predefined schemas.
Schema-on-write for data optimization.
Optimized for batch processing and reporting.
Ideal for business intelligence and structured data analysis.
Typically higher cost due to data structuring.

Key Difference:

Data Lake is best for diverse, raw data, while Data Warehouse is for structured data, with both often used together to meet various data storage and analysis needs.

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Data lake
vs
Data Warehouse

Data Lake	BASIS	Data Warehouse
Raw, Structured, Semi-Structured	DATA STRUCTURE	Processed , Structured
Not Determined	PURPOSE SERVED	In use
Data Scientists	USERS	Business Professionals
HIgh Accessibility & Quik to change	ACCESSIBILITY	Complicated & Costly to make changes
Low-cost storage	STORAGE	Expensive for large volumes of data
Maturing	SECURITY	Mature
Schema-on-read	PROCESSING	Schema-on-write
Highly agile & configurable	AGILITY	Less agile & fixed configuration
Decoupled	STORAGE	Tightly coupled

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Qustions ??
Thank You

The next Lecture will be :

Introduction to Data Warehouse

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